

for Iteration Statement

ELEC1006: ENGINEERING COMPUTING

Iteration Statement (loops)

- Loops are used to repeat a statement a certain number of times when a condition is met.
- There are three types of loops:
 - *while* loop.
 - *do-while* loop.
 - *for* loop. ←

The *for* Loop

- *for* loop is used when we know how many iterations (loops) are required.
- Test each expression before each iteration in the *for* loop and so, it is known as a pre-test loop similar to *while* loop.
 - Note: *do-while* loop is known as a post-test loop, where the statements executed at least once (one iteration) before the expression is checked.

for Loop Syntax

- The format is:

```
for (initialisation; expression; increase or decrease)
{
    statements
}
```

Diagram illustrating the components of a *for* loop header:

- for* (initialisation; **expression**; **increase or decrease**)
- Test** (points to **expression**)
- Update** (points to **increase or decrease**)

– Note: No ; after the)

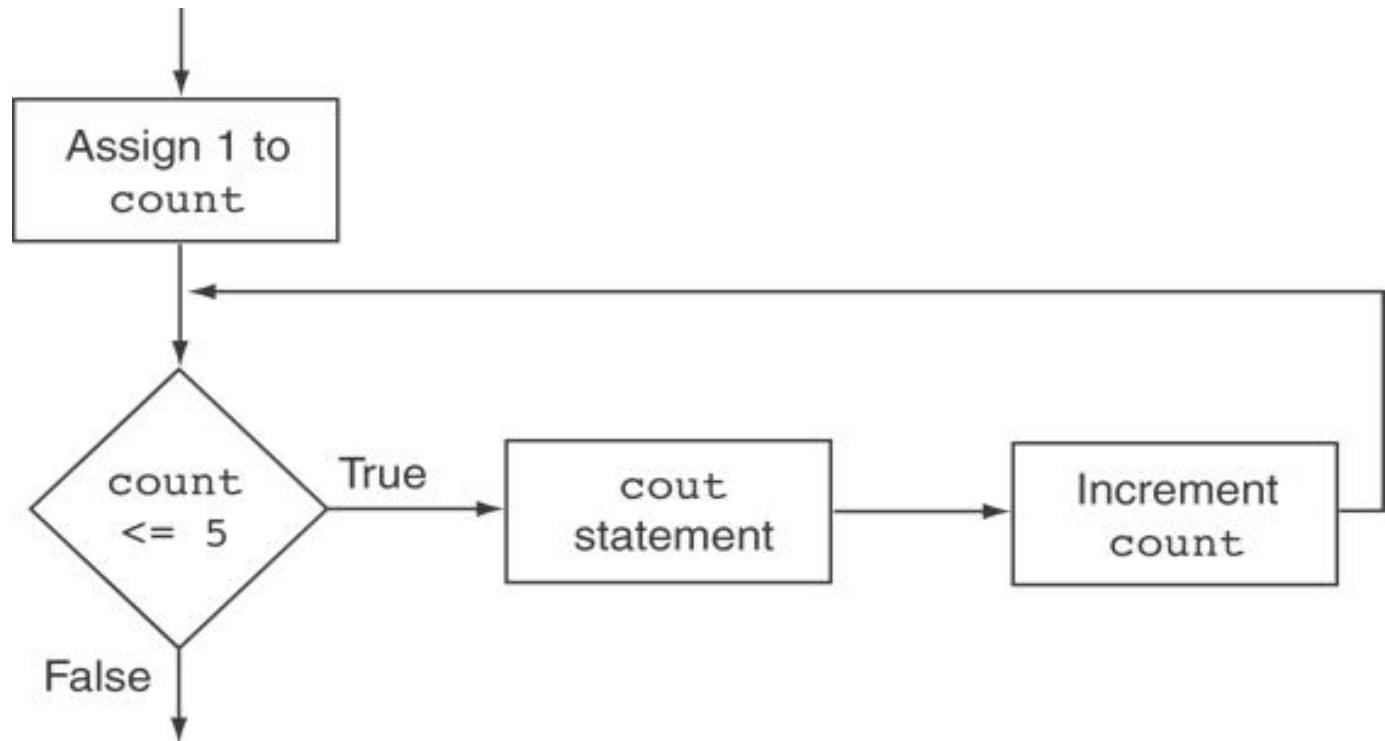
The mechanics:

- Perform initialisation
- Evaluate test expression
 - If true, execute statement
 - If false, terminate loop execution
- Execute update, then re-evaluate test expression

Example

```
for (int count = 1; count <= 5; count++)  
{  
    cout << "Hello" << endl;  
}
```

Flowchart



while Loop Equivalent

```
for (int count = 1; count <= 5; count++)  
{  
    cout << "Hello" << endl;  
}
```

for loop is like an advanced *while* loop

```
int count=1;  
while (count <= 5)  
{  
    cout << "Hello" << endl;  
    count++;  
}
```

Variations using commas

- Can also have multiple **initialisation** and **update** expressions. Separate the statements with **commas** , :

```
int x, y;
for (x=1, y=1; x <= 5; x++, y++)
{
    cout << x << " plus " << y
        << " equals " << (x+y)
        << endl;
}
```

Initialisations

updates

Scope of Variables

- You can declare variables in the **initialisation** expression.
- Variable `i` only exist in the `for` loop `{ ... }`, not outside of it.

```
int sum = 0;
for (int i = 0; i <= 10; i++)
{
    sum += i;           // sum = sum + i
}
```


More info

- [1] cplusplus.com: Statements and flow control
<https://cplusplus.com/doc/tutorial/control/>
- [2] learncpp.com: 5.7 – For statements
<https://www.learncpp.com/cpp-tutorial/57-for-statements/>